

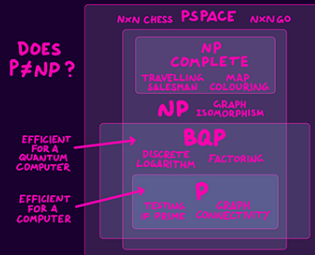
Artificial Intelligence



Dé Céadaoin, 2ú Márta

MAP OF COMPUTER SCIENCE

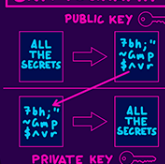
COMPUTATIONAL COMPLEXITY



INFORMATION THEORY

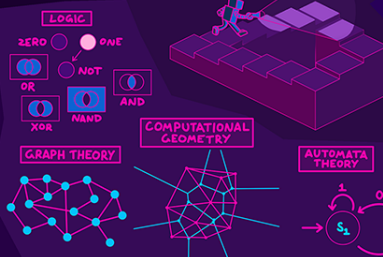


CRYPTOGRAPHY



THEORETICAL COMPUTER SCIENCE

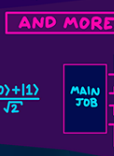
COMPUTABILITY THEORY



TURING MACHINE



AND MORE



ALGORITHMS

BUBBLESORT(A)

- 1: GO FROM LEFT TO RIGHT.
- 2: COMPARE EACH PAIR.
- 3: IF LEFT ONE HIGHER, SWITCH.
- 4: DO UNTIL NO MORE SWITCHES.

BUBBLE SORT $O(n^2)$

1 2 3 4 5 6 7 8

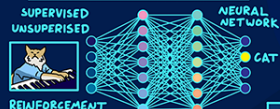
MERGE SORT $O(n \log n)$

1 2 3 4 5 6 7 8

ANALYSIS OF ALGORITHMS

ALGORITHMIC COMPLEXITY

MACHINE LEARNING



COMPUTER VISION



IMAGE PROCESSING

OPTIMISATION



ARTIFICIAL INTELLIGENCE



BOOLEAN SATISFIABILITY

x_1 OR x_2 OR $\bar{x}_3$ (SAT)

$\bar{x}_1$ OR $\bar{x}_2$ OR x_3

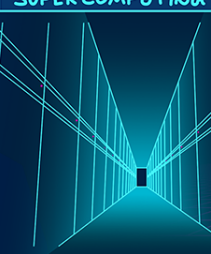
$\bar{x}_1$ OR x_2 OR $\bar{x}_3$

$\bar{x}_1$ OR x_2 OR x_3

ROBOTICS

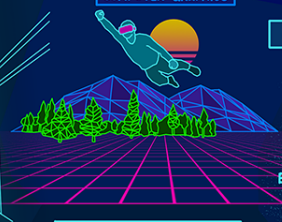


SUPER COMPUTING



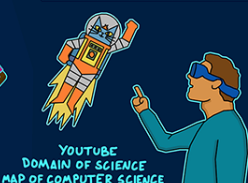
APPLICATIONS

COMPUTER GRAPHICS



VIRTUAL REALITY

AUGMENTED REALITY

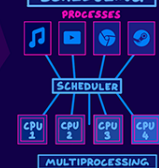


HUMAN COMPUTER INTERACTION

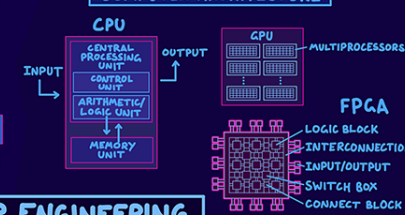


INTERNET OF THINGS

SCHEDULING



COMPUTER ARCHITECTURE



COMPUTER ENGINEERING

SOFTWARE AND PROGRAMMING LANGUAGES



OPERATING SYSTEMS



COMPUTATIONAL SCIENCE

COMPUTATIONAL PHYSICS, NUMERICAL ANALYSIS

HACKING

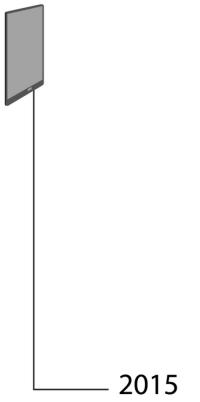
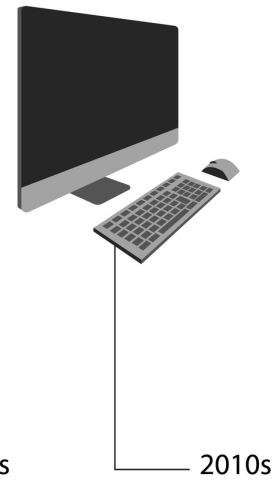
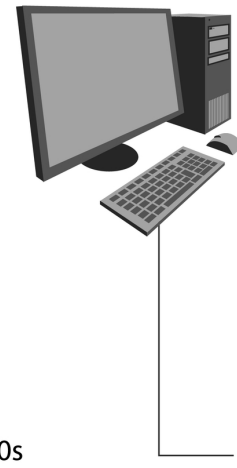
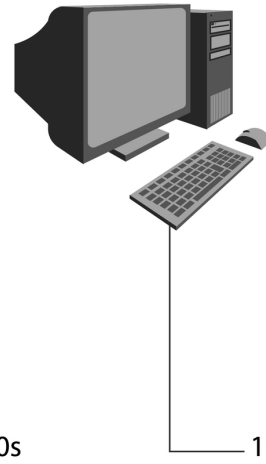
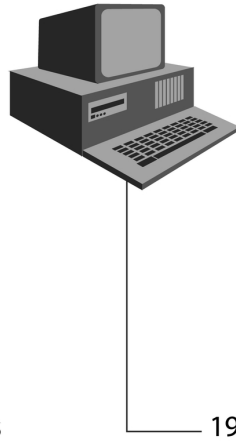
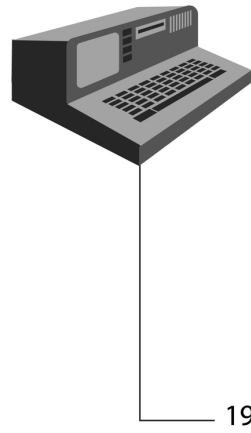
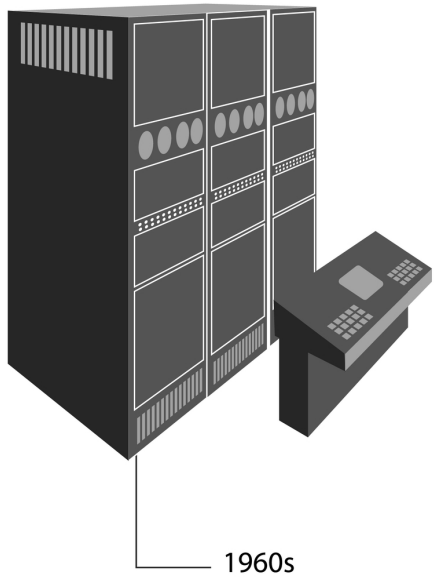


BIG DATA



BY DOMINIC WALLIMAN ©2017

Evolution of computers



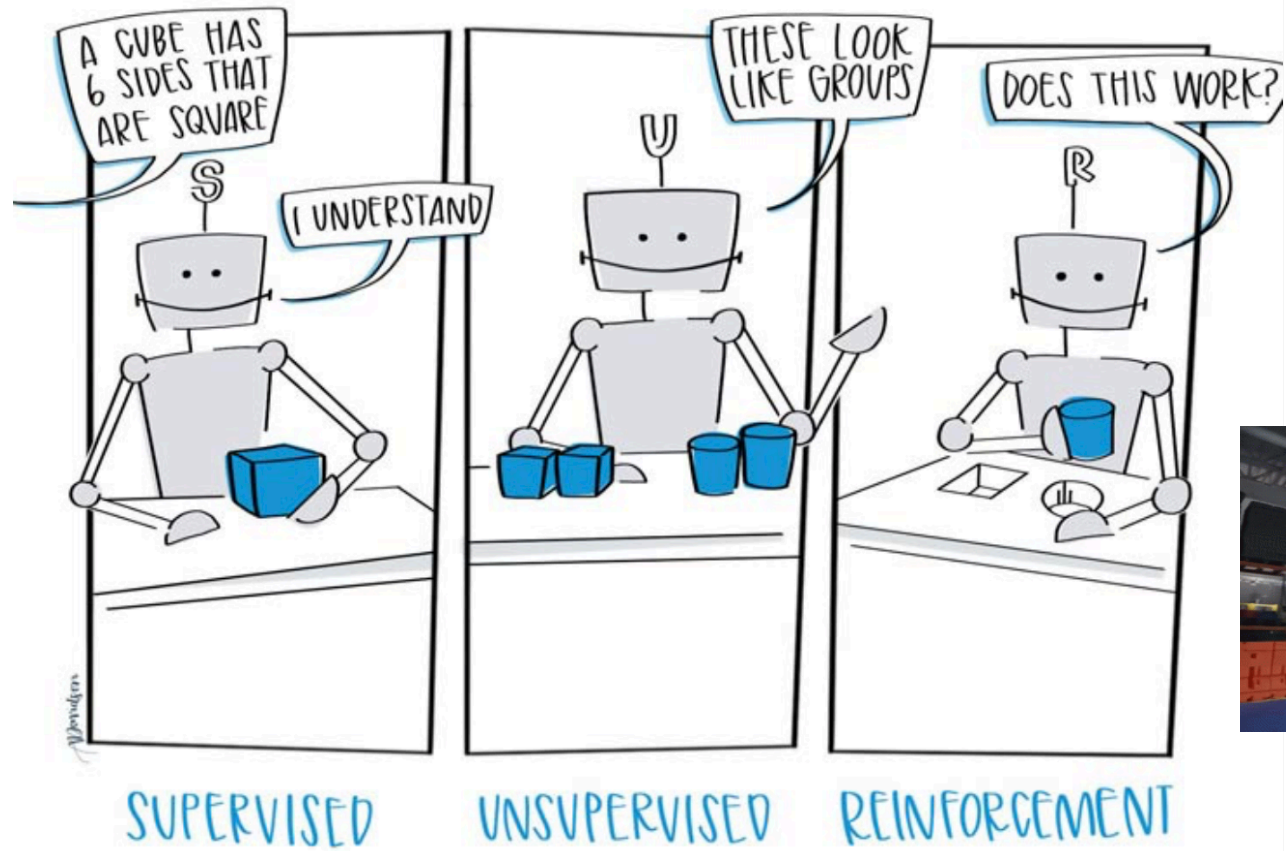
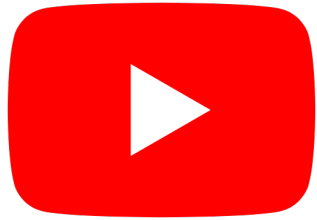
Intelligence?

Morning Routine

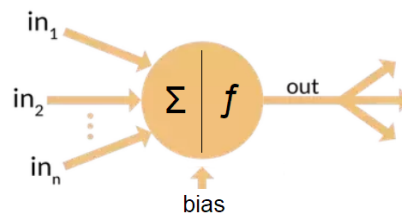
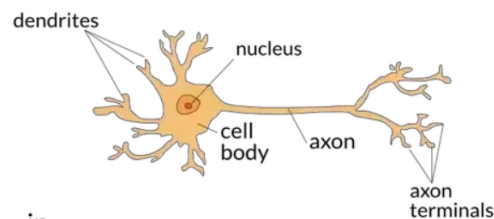
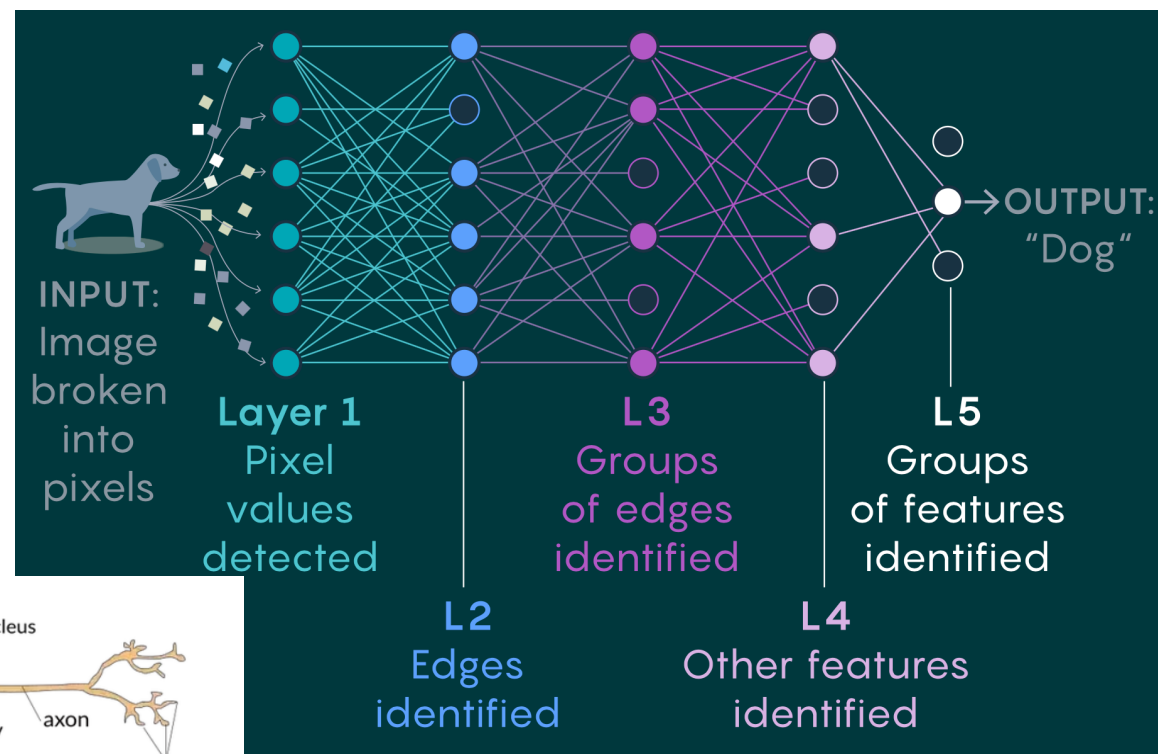
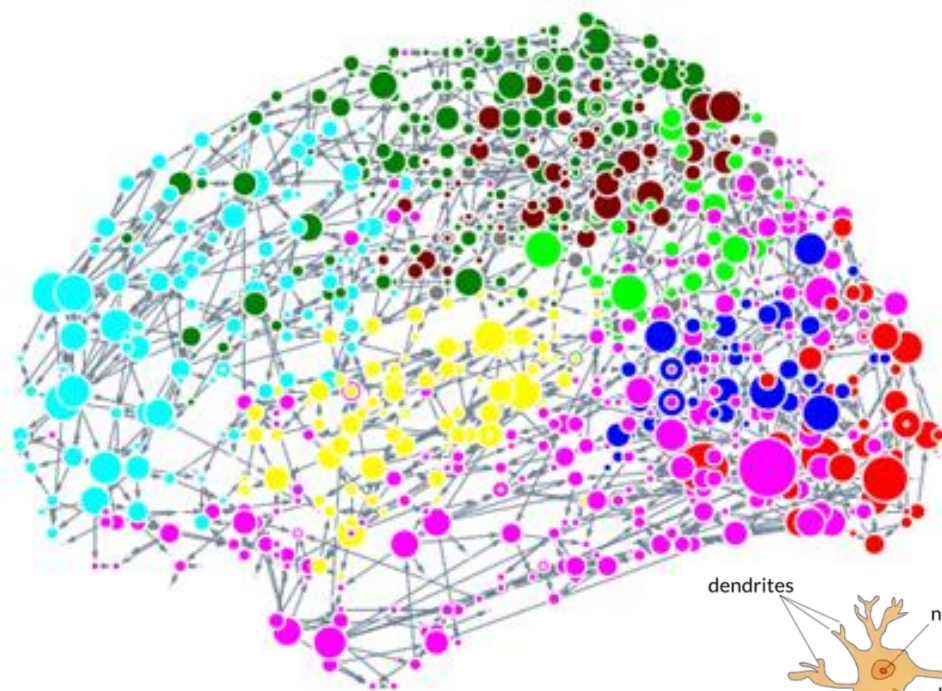


- Walk down stairs
 - Find door
 - Enter kitchen
 - Locate table
 - Sit down
- Charge battery

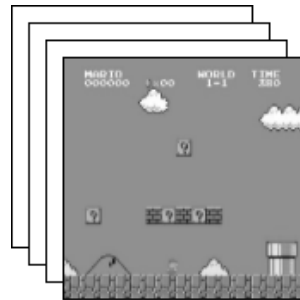
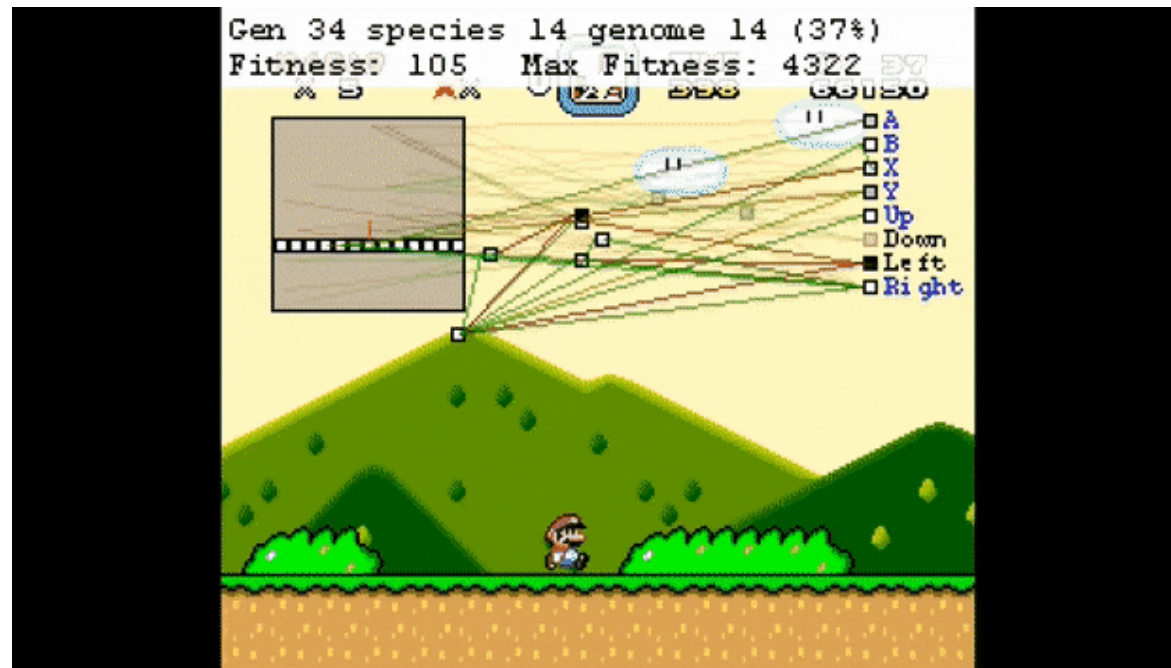
MACHINE LEARNING



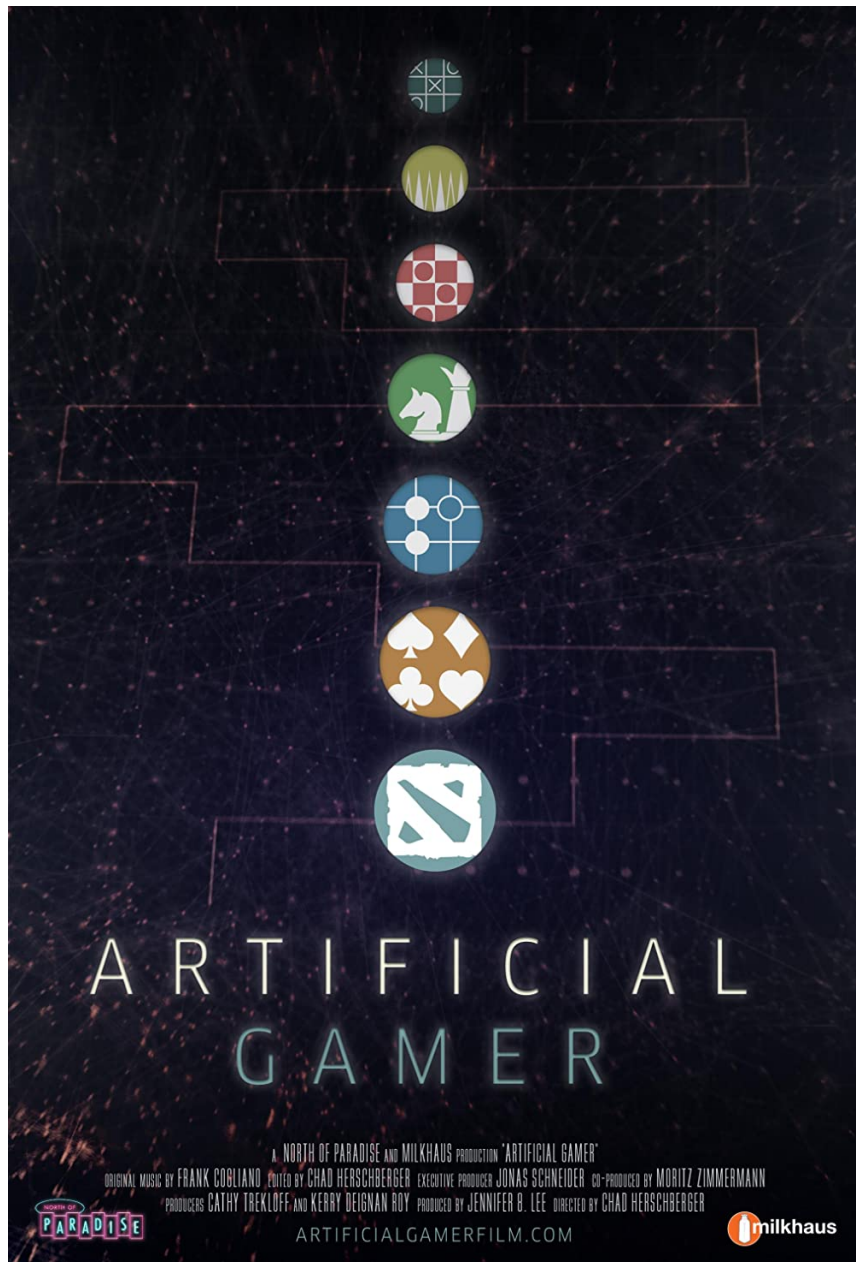
Neural Networks



Deep Learning: Try and Try Again



```
[[140 140 140 ... 140 140 140]]  
[1][[140 140 140 ... 140 140 140]]  
[1][1][[140 140 140 ... 140 140 140]]  
.. [1][1][[140 140 140 ... 140 140 140]]  
[1].. [1][140 140 140 ... 140 140 140]]  
[1][1].. [140 140 140 ... 140 140 140]]  
[1][1][1]...  
[1][1][142 124 115 ... 89 147 83]  
[1][132 94 86 ... 109 126 83]  
[114 99 120 ... 113 81 54]]
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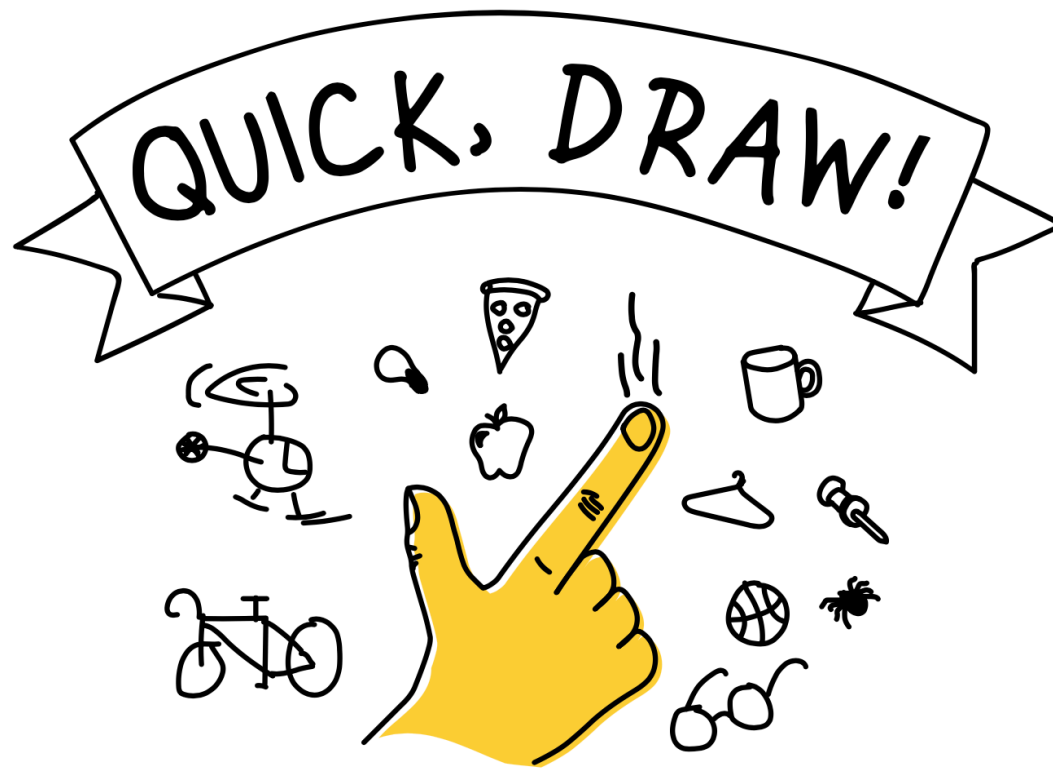


SAP Conversational AI



SAP AI Business Services





Can a neural network learn to recognize doodling?

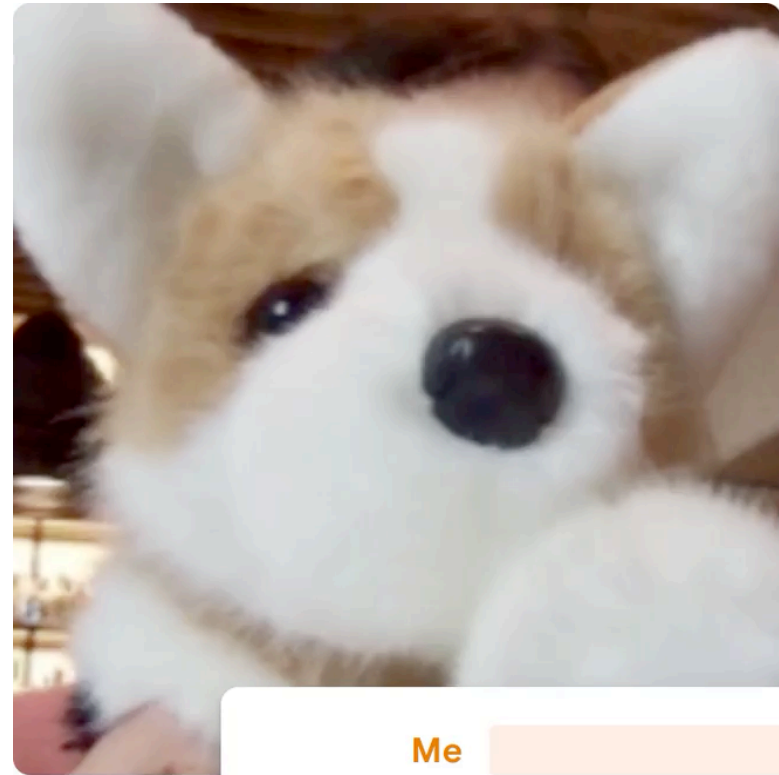
Help teach it by adding your drawings to the [world's largest doodling data set](#), shared publicly to help with machine learning research.

Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started



Me



Me + Dog <3

20%

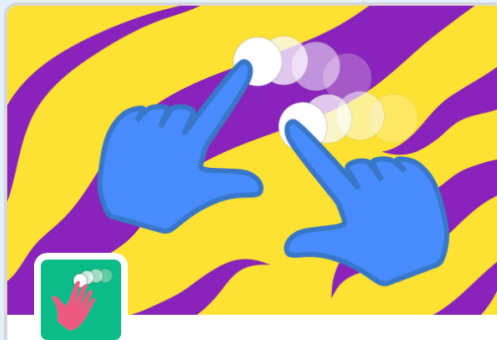


Scratch AI Blocks



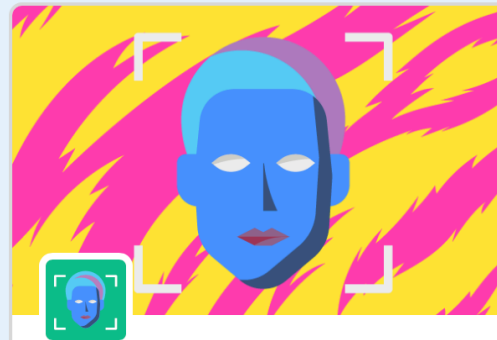
Teachable Machine

Use Google Teachable Machine models in your Scratch project.



Hand Sensing

Sense hand movement with the camera.



Face Sensing

Sense face movement with the camera.



Body Sensing

Sense body position with the camera.